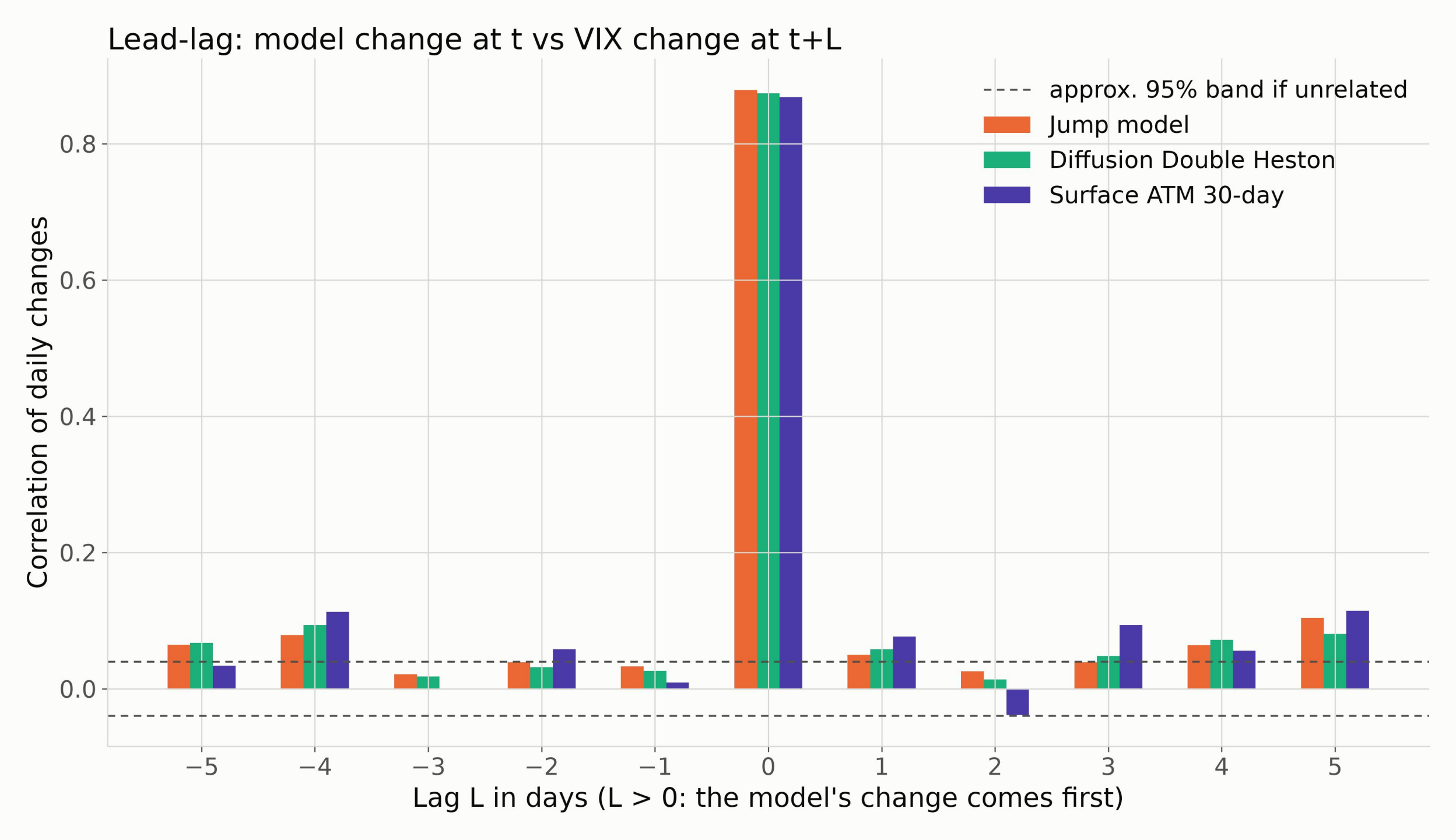

Lead-lag: model change at t vs VIX change at t+L
approx. 95% band if unrelated
Jump model
Diffusion Double Heston
Surface ATM 30-day
Correlation of daily changes
0.8
0.6
0.4
0.2
0.0
−5
−4
−3
−2
−1
0
1
2
3
4
5
Lag L in days (L > 0: the model's change comes first)